

Blueprint Sistem ML Swing Trading IDX (XGBoost) — 5D Swing

Tujuan: Membangun pipeline rekomendasi harian saham IDX berbasis ML (XGBoost) untuk horizon 5 hari dengan pendekatan technical + bandarmology, siap dipakai sebagai decision support dan bisa di-retrain harian.

1) Objective & Guardrails

- **Target:** Profit window 5 hari kerja, hit-rate $\geq 55\%$ pada top-10 picks, expectancy $> 0.3R$.
- **Universe:** Saham IDX likuid (filter: Avg Turnover $\geq$ Rp1 miliar/hari 20D, Free float OK). Exclude: suspensi, fraksi sangat kecil.
- **Latency:** Update setelah market close (EOD), rekomendasi dipakai next day.
- **Risk:** Per posisi 0.5–1.0% ekuitas; max 10–15 posisi.
- **Cost model:** Biaya beli+jual 0.3–0.5% total (d disesuaikan broker).

2) Data & Ingestion

Sumber: GoAPI (OHLCV harian, broker summary/akumulasi), Corporate actions (dividen, stock split), IDX calendar (libur, suspensi).

Skema data mentah (parquet/csv) - `prices`: date, ticker, open, high, low, close, volume, value, adj_close - `brokers`: date, ticker, buy_val_top5, sell_val_top5, net_val_top5, dealer_net, foreign_net, buyer_conc_idx, seller_conc_idx - `meta`: ticker, sector, subsector, free_float, shares_out, first_listed - `calendar`: date, is_trading_day

Ingestion harian (EOD) 1) Tarik OHLCV, broker summary, meta 2) *Sanity checks*: missing/dup date, outliers (gap $> 40\%$ tanpa CA), volume=0 3) Simpan ke `datalake/raw/YYYY-MM-DD/`

3) Feature Engineering (± 30 – 60 fitur)

A. Trend & Momentum - MA: SMA/EMA(5,10,20,50,100,200), slope MA, price/MA ratios - MACD: macd, signal, hist, delta hist 3D - RSI(14), RSI(2), Stoch %K/%D, MFI(14) - Rate of Change (ROC 5/10), CCI(20)

B. Volatilitas & Range - ATR(14), True Range %, HV(20), candle body% & upper/lower shadow%

C. Volume & Partisipasi - Volume z-score (20D), volume/avg20, OBV change, VWAP gap%

D. Price Action & Breakout Flags - HH/HL status (structural trend), inside/outside bar flags, 20D/55D breakout up/down, gap up/down %

E. Bandarmology (proxy) - Top5 buyer/seller net value%, concentration index (Herfindahl proxy), dealer net sign, foreign net% (jika tersedia) - Accum/Distrib day count (5D/20D)

F. Microstructure & Likuiditas - Turnover 20D, median spread proxy (close-high/low), lot-to-value ratio

G. Seasonality/Calendar - DayOfWeek, daya ke-n dalam bulan, pra/pasca dividen ($\pm 10D$), akhir/buka bulan flags

Target/Label - Klasifikasi: $y = 1$ jika $\frac{\max(\text{Close}[+1\dots+5])}{\text{Close}[0]} - 1 \geq \text{TP\%}$ sebelum kena SL% ($\frac{\min(\text{Close}[+1\dots+5])}{\text{Close}[0]} - 1 \leq -\text{SL\%}$). - Default: TP = +3%, SL = -2% (bisa dikalibrasi). - **Regresi (opsional):** 5D forward return $\frac{\text{Close}[+5]}{\text{Close}[0]} - 1$.

Leakage guards - Semua fitur hanya pakai data $\leq t$ (no look-ahead) - Gunakan `adj_close` setelah corporate actions

4) Modeling Pipeline (XGBoost)

Baseline: XGBoostClassifier untuk `y` (profit-before-loss dalam 5D)

Preprocessing - Imputasi missing (median/0), winsorize outliers (1/99 pct), standardize *optional* (tree-based umumnya tak perlu)

Training scheme (Walk-Forward) - Split tahunan/kuartalan: contoh 2019–2022 train → 2023 val; geser jendela → 2020–2023 train → 2024 val - Atau rolling 24M train → next 3M test (berulang)

Class imbalance - Gunakan `scale_pos_weight = (#neg/#pos)`

Hyperparams (starting point) - `n_estimators`: 500–1500 - `max_depth`: 4–7 - `learning_rate`: 0.03–0.1 - `subsample`: 0.7–0.9 - `colsample_bytree`: 0.6–0.9 - `reg_alpha`: $1e-3$ – $1e-1$, `reg_lambda`: 1–5

Calibration - Platt/Isotonic untuk probabilitas; stabil untuk ranking picks

Explainability - Feature importance (gain/cover), SHAP (top features per ticker)

5) Backtest & Execution Rules

Backtest (event-driven, next-day open/close) - Sinyal dievaluasi EOD $t \rightarrow$ eksekusi di open $t+1$ (atau VWAP $t+1$) - Entry: pilih `N` saham dengan `p>=threshold` (misal top-10), exclude duplikasi sektor berlebih bila perlu - Exit: otomatis TP/SL intraperiod (simulasi intraday menggunakan H/L harian) atau waktu 5D habis - Biaya transaksi dan slippage dimasukkan

Metrics - Hit-rate, Avg win/loss, PF, Sharpe (daily), MaxDD, Turnover, Kelly fraction (informasi) - Per-ticker/per-sektor drilldown

Live Execution (harian) 1) 16:30–18:00: Ingest + feature compute + scoring (model terkini) 2) 18:00: Ranking + filter risk (likuiditas, overlap sektor) 3) 18:05: Generate **Trade Plan** (ticker, entry ref, TP/SL, size) 4) Next day open: eksekusi manual/auto; log order 5) Monitoring harian: move to breakeven (opsional), partial take profit (opsional)

6) Risk & Portfolio Management

- **Sizing:** Fixed fractional (0.5–1% risk) → $\text{shares} = (\text{risk} * \text{equity}) / (\text{entry} - \text{stop})$
 - **Cap exposure:** max 15 posisi, max 20% sektor, max 2 posisi per emiten (bila multi-setup)
 - **Kill switch:** jika 10 trade berturut rugi atau MaxDD > X%, turunkan sizing 50% atau pause
 - **Regime filter:** IHSG di atas MA200 & Breadth OK untuk full risk; jika di bawah → kurangi N & sizing
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7) MLOps & Monitoring

- **Retrain:** harian/2–3 hari sekali (incremental) + bulanan full retrain
 - **Model registry:** simpan versi model, hash data, metrik OOS
 - **Drift:** pantau distribusi fitur/label, metrik live vs backtest
 - **Alert:** jika hit-rate 20 trade terakhir < p5 backtest → kirim notifikasi
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8) Output & Integrasi

- **Artefak harian**
 - `reco_YYYYMMDD.csv`: ticker, prob, exp_return, entry, TP, SL, size
 - `watchlist_YYYYMMDD.abk` (opsional export ke Amibroker) / `csv`
 - `explain_YYYYMMDD.html`: SHAP summary
 - **Dashboard** (opsional): Streamlit/Gradio untuk 3 klik workflow: *Ingest* → *Score* → *Export Plan*
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9) Folder Structure (contoh)

```
project/
  config/
    params.yaml
  data/
    raw/           # goapi dumps
    interim/       # cleaned
    features/      # feature matrices
    models/        # xgb.bin + meta
    reports/       # backtest, explain
  src/
    ingest.py
    features.py
    label.py
    train.py
    score.py
    backtest.py
    risk.py
    utils.py
  notebooks/
```

10) Konfigurasi (params.yaml contoh)

```
universe:
  min_turnover_idr: 1000000000
  min_price: 100
  exclude_suspended: true
label:
  horizon_days: 5
  take_profit: 0.03
  stop_loss: 0.02
model:
  algo: xgboost
  n_estimators: 800
  max_depth: 6
  learning_rate: 0.05
  subsample: 0.8
  colsample_bytree: 0.7
risk:
  risk_per_trade: 0.0075
  max_positions: 12
  max_sector_weight: 0.2
execution:
  entry_price: next_open
  exit_logic: tp_sl_or_timeout
```

11) Pseudocode Pipeline (ringkas)

```
# 1) Load & Clean
df = load_prices(); br = load_brokers(); meta = load_meta()
df = clean(df); br = clean(br)
# 2) Features
X = make_features(df, br, meta)
# 3) Label
y = make_labels(df, tp=0.03, sl=0.02, horizon=5)
# 4) Split walk-forward
splits = rolling_splits(X, y, lookback_months=24, test_months=3)
# 5) Train & validate
for tr, te in splits:
    model = XGBClassifier(**params).fit(X[tr], y[tr])
    eval_metrics(model, X[te], y[te])
# 6) Score EOD
proba = model.predict_proba(X_live)[: ,1]
ranked = rank_filter(proba, liquidity, regime)
# 7) Sizing & Plan
```

```
plan = size_positions(ranked, risk_per_trade, tp, sl)
export(plan)
```

12) Backtest Rules Detail (5D swing)

- Entry di **Open T+1** setelah sinyal EOD T; gunakan `Close T` sebagai basis label.
- TP/SL dievaluasi menggunakan **High/Low** harian: jika **Low** $\leq$ **Stop** lebih dulu $\rightarrow$ hit SL; jika **High** $\geq$ **TP** lebih dulu $\rightarrow$ hit TP; jika tidak, keluar di **Close T+5**.
- Slippage default 5–15 bps; biaya 30–50 bps round-trip.
- Reinvesting ekuitas dinamis; cash drag diperhitungkan jika tidak ter-isi semua.

13) Evaluasi & Thresholding

- Tentukan **threshold probabilitas** dengan ROC/PR curve $\rightarrow$ pilih titik yang memaksimalkan `Profit Factor` OOS, bukan sekadar AUC.
- Lakukan **calibration** supaya probabilitas $\sim$ reliabel $\rightarrow$ memudahkan **position sizing adaptif** (besar lot $\propto$ edge).

14) Integrasi Bandarmology (Praktikal)

- Fitur inti: `net_val_top5/turnover`, `buyer_conc_idx`, `dealer_net_sign`, `foreign_net_ratio`, `accum_day_5/20`.
- Normalisasi per ticker (z-score rolling) agar model belajar **anomali** relatif historis emiten tsb (bukan absolut lintas saham).
- *Ablation test*: cek dampak menyalakan/mematikan set fitur bandar pada hasil.

15) Governance & Safety

- **Data change log**: simpan snapshot harian agar reproducible.
- **Model cards**: catat tanggal, data range, metrik, catatan risiko.
- **Compliance**: hindari front-running isu sensitif; transparansi ke diri sendiri lewat log.

16) Roadmap 4 Minggu (versi cepat)

M1: Ingest GoAPI + pembersihan + 30 fitur teknikal dasar

M2: Labeling 5D TP/SL + XGBoost baseline + walk-forward + backtest v1

M3: Tambah fitur bandarmology + regime filter + sizing + dashboard simple (Streamlit)

M4: Calibration + SHAP + alerting + dokumentasi + SOP eksekusi

17) SOP Harian (3 klik)

1) **Sync Data** (Ingest → Build features) 2) **Score & Rank** (Apply model → Top N) 3) **Export Trade Plan** (CSV/Amibroker watchlist) + review manual 5 menit

18) Checklist Kualitas (sebelum live)

- ☐ Backtest OOS ≥ 2 tahun
- ☐ Top-10 hit-rate $\geq 55\%$, PF ≥ 1.3 , MaxDD wajar
- ☐ Sensitivitas biaya/slippage lulus
- ☐ Robust terhadap removal 10% fitur (tidak runtuh)
- ☐ Dokumentasi + repeatability OK